

Piyush Daryanani

250-981-0368 | daryanani@unbc.ca | [linkedin.com/in/piyushdaryanani](https://www.linkedin.com/in/piyushdaryanani)

EDUCATION

B.Sc. Computer Science | University of Northern British Columbia

Expected April 2027

- **Dean's List** (Top 10% GPA), Fall 2025 & Winter 2026
- **Relevant Coursework:** Data Structures & Algorithms I & II, Database Systems, Software Engineering, Operating Systems, Computer Architecture, Artificial Intelligence, Computer Networks, Discrete Mathematics

SKILLS

Languages: Python, Java, C++, C, JavaScript, TypeScript, Bash

Frontend & Backend: Angular, HTML/CSS, Node.js, Express.js, Django, REST APIs, JSON, object-oriented design

AI & Data: TensorFlow.js, COCO-SSD, LLM integration, computer vision (object & ArUco detection), log analysis

Databases: MySQL, PostgreSQL, relational schema & E-R design, normalization

Tools & Systems: Git, GitHub, Linux, ROS 2, Raspberry Pi, AWS (EC2), 3D printing

EXPERIENCE

Software Developer & Research Assistant, Social Robotics Lab | University of Northern British Columbia

Feb 2025 – Present

- Design and build a **screen-free, hardware-based interactive game platform** integrated with social robots for a child-focused learning research study
- Run **prototype testing and pilot studies with children**, then analyze feedback and implement iterative design and interaction changes
- Develop and maintain **Python / ROS-based applications, LLM integrations**, and data-processing tools supporting multi-robot research deployments
- Build **automated data checks** and analyze **system logs** to ensure data consistency and diagnose runtime issues in **Linux-based environments**

Department Supervisor & Machine Trainer | The Home Depot

Feb 2024 – Present (Prince George, BC)

- Lead and develop a team of **15–20 associates** — coaching, scheduling, and performance feedback to drive customer service and top-line sales
- Own **stock ordering and inventory management**, resolving discrepancies and organizing overhead to keep the department consistently in-stock
- Investigate and resolve **escalated customer issues**, and communicate sales goals, promotions, and safety / asset-protection standards in weekly department meetings
- As **Machine Trainer**, onboard and certify associates on equipment operation, process compliance, and safety standards

PUBLICATIONS

Kupaei, R., **Daryanani, P.**, Shaju, A., Olaniyan, M. A., Chavez-Castaneda, G., & Chandra, S. (2026). “Social Robots on the Loose: Supporting Children’s Independent Learning in the Wild.” **Proceedings of the 21st ACM/IEEE International Conference on Human–Robot Interaction (HRI ’26)** — Full Paper (23.2% acceptance rate). doi.org/10.1145/3757279.3785580

PROJECTS

Screen-Free Robot-Integrated Learning Games | Raspberry Pi, Computer Vision, LLMs, 3D Printing

May 2026 – Present

- Built **5 fully hardware-based games** — Memory Matrix, Match Point, Scramble Word, Misty Maze, and Tangram — on Raspberry Pi with touch, RFID, buzzers, AI object detection, and ArUco markers; custom parts **3D-printed**, with no screens
- Integrated **LLMs and social robots** (QT, Reachy-mini, Misty); implementation complete and now running **unsupervised studies** where children play independently, with data collected toward a planned publication

QT Robot — Autonomous Interaction System | Python, Linux, ROS 2 (Active Minds Summer Camp)

May 2025 – Aug 2025

- Developed **Python-based interaction and control software** for the QT humanoid robot, integrating web-based educational games on a **Linux (ROS 2)** platform
- Implemented and refined **autonomous robot behaviors** and interaction flows, enabling fully **unsupervised operation** in real classroom summer-camp settings (no adult control)
- Deployed across **7 summer camps with 113 child participants**; collected and processed interaction logs with structured quality checks to evaluate reliability, engagement, and task completion

Task Reordering in Sparse Dependency Matrices | Java, Algorithms, Graph Theory

Sep 2025 – Dec 2025

- Optimized task scheduling in **large-scale software systems** by minimizing feedback dependencies in sparse Dependency Structure Matrices (DSMs)
- Implemented **Kosaraju’s and Tarjan’s algorithms** to detect Strongly Connected Components, build meta-graphs, and apply topological sorting; evaluated on **27,000+ tasks**

Nature’s Heroes — Object Detection Game | JavaScript, TensorFlow.js, COCO-SSD

Jan 2025 – Apr 2025

- Integrated **TensorFlow.js with COCO-SSD** for real-time, in-browser object detection powering an educational waste-classification game
- Built **client-side ML inference** with state-driven scoring; system was used in the peer-reviewed HRI 2026 study above

OCTBS — Online Cinema Ticket Booking System | Angular, Node.js, Express, MySQL

Sep 2025 – Dec 2025

- Built a full-stack booking app (**Angular, Node.js/Express, MySQL**) with real-time seat availability, an admin management dashboard, and automated confirmation emails
- Implemented **transaction-safe seat-locking logic** and **RESTful APIs** to prevent race conditions and double bookings under concurrent access